RANSOMWARE DETECTION SIMULATOR

Faculty Presentation & Viva Defense Master Guide — B.Tech CSE (AI/ML) Major Project

Candidate Degree: B.Tech CSE (AI/ML)	**Academic Track:** Defensive Cybersecurity & EDR
Application: Web-based SOC Simulator	**Core Stack:** Python, Flask, Watchdog, Scikit-learn, SQLite

1. The 60-Second Opening Pitch (What to Say First)

When your faculty asks *'Introduce your project and explain what problem it solves,'* deliver this exact pitch with confidence:

Exact Script to Speak:
*"Respected faculty members, my major project is the **Ransomware Detection Simulator**. Traditional antivirus software relies on static file signatures — such as MD5 or SHA-256 hashes. However, modern cyber adversaries easily defeat signature detection by recompiling their payloads, using polymorphic crypters, or launching zero-day ransomware where no signature exists.*

*To solve this problem, my project implements **real-time behavioral anomaly detection** — the technology powering modern Next-Generation Antivirus (NGAV) and Endpoint Detection & Response (EDR) systems. The system monitors operating system file activity using Python Watchdog, tracks event velocity inside a sliding time window, detects rapid file-renaming and ransomware extension changes (such as .locked or .crypto), and computes a transparent, explainable threat score from 0 to 100.*

*Furthermore, it couples rule-based heuristics with a **Scikit-Learn Random Forest machine learning classifier** to evaluate behavioral telemetry, and demonstrates automated defensive containment in a completely safe, sandbox-isolated environment without endangering host system files."*

2. System Architecture & Component Breakdown

When your teacher asks how the system is designed under the hood, break it down into these 5 clear modular layers:

Component Layer	Key Technologies	Engineering Responsibility
1. Presentation / SOC Dashboard	HTML5, Bootstrap 5, Chart.js 4.4, JavaScript ES6	Renders dark-themed SOC UI, live KPI cards, timeline charts, incident tables, and speed sliders.
2. Backend Web Server	Python, Flask 3.1, REST APIs	Coordinates session authentication, state machine controls, telemetry polling, and CSV export.
3. Safe Simulation Engine	Thread state machine, Synthetic payloads	Executes harmless dummy file operations inside isolated demo_data/sandbox/ with path traversal guards.
4. Detection & Scoring Pipeline	Sliding Window, Heuristic Rules, Random Forest	Maintains a sliding deque of events, computes velocity, calculates explainable threat score, triggers alerts.

5. Persistence & Forensics	SQLite 3 (PRAGMA foreign_keys)	Stores raw security events, persistent security incidents, audit logs, and dynamic detection thresholds.

3. Step-by-Step Live Demonstration Walkthrough

Follow this exact sequence on your laptop during the faculty demonstration to ensure a smooth, professional presentation:

Step 1: Login & Accessing the SOC

Launch http://127.0.0.1:5000 in your browser.
Click the convenient **Admin (Lead SOC)** 1-click button on the login screen.
Say: *'We authenticate into the security operations center. The platform supports role-based access for Lead SOC Administrators and Security Analysts.'*

Step 2: Tour the SOC Dashboard

Point to the 4 KPI cards (Sandbox Files, Safe Operations, Suspicious Events, Threats Detected).
Show the live **Threat Timeline** (Chart.js) and **Threat Severity Doughnut**.
Say: *'This dashboard gives security analysts instant situational awareness. Currently, the environment is at a benign baseline.'*

Step 3: Run Benign Simulation (Negative Test / Baseline)

Click **Simulation Engine** in the sidebar. Select **Benign User Activity** and click **Start**.
Watch the live terminal feed and sandbox file list create dummy files (annual_budget.xlsx.txt, research_paper.docx.txt).
Say: *'Here we simulate normal human activity: slow typing, standard edits, and safe renaming. Notice that the threat score remains in the Low tier (under 30), and zero false alerts are generated.'*

Step 4: Run Ransomware Burst Attack (Positive Test / Anomaly)

On the Simulation page, select **Ransomware Burst Attack** and click **Start**.
Observe the rapid overwriting of dummy files with simulated cipher text, batch renaming to .locked, and the dropping of a simulated ransom note (README_RESTORE_FILES_DEMO.txt).
Switch to the Dashboard: Point out the red **CRITICAL THREAT DETECTED** banner that instantly appears.
Say: *'When an automated encryption burst occurs, our sliding window detects high velocity and extension tampering. The threat score spikes to 89/100 (Critical), and our automated defense suspended the process to prevent further spread.'*

Step 5: Explain the Mathematical Attribution

Click **Incident History**, then click **Details** on the top incident.
Show the faculty the exact mathematical formula string and the Scikit-learn feature vector modal.
Say: *'Our system is completely explainable. Rather than a black-box decision, the modal displays the exact point breakdown: 14 points from rename rate, 40 points from extension tampering, 25 points from modification bursts, and 10 points from event velocity.'*

Step 6: Show File Monitoring & Safety Bounds

Click **File Monitoring**. Try entering C:\Windows and show that the path validator immediately blocks it.
Click **Sandbox** and click **Start Monitoring** to stream live OS Watchdog events.
Say: *'Our watchdog observer captures live operating system I/O events without ever reading private file contents, ensuring zero host overhead and total privacy.'*

Step 7: Download CSV Forensic Report

Go to **Reports & Export** and click **Download Incidents CSV**.
Say: *'Finally, our platform exports comprehensive audit telemetry conforming to NIST SP 800-61 incident response guidelines for forensic analysis.'*

4. The Threat Scoring Mathematical Formulation

Faculty love mathematical formulas. Explain this transparent calculation clearly:

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Score Formula:
Score = min(100, S_rename + S_extension + S_burst + S_velocity + S_ml)
```

Component	Max Weight	Mathematical Formulation	Detection Focus
Rename Score (S_rename)	35 pts	min(35.0, (N_renames / Threshold_rename) × 35.0)	Detects mass automated file renaming within the sliding time window.
Extension Score (S_ext)	40 pts	0.7 × W_ext (1 ext) or 1.0 × W_ext (≥2 exts)	Detects known ransomware extensions (.locked, .crypto, .enc, .wnry).
Burst Score (S_burst)	25 pts	min(25.0, (N_mods / Threshold_mod) × 25.0)	Detects rapid overwriting of file content (in-place cipher writes).
Velocity Bonus (S_velocity)	10 pts	min(10.0, (Velocity / Threshold_velocity) × 10.0)	Rewards high operations-per-second exceeding human baseline.
ML Bonus (S_ml)	15 pts	+15 if Random Forest confidence ≥ 75% on borderline cases	AI-driven confirmation for ambiguous behavioral signals.

5. AI/Machine Learning Integration (Scikit-Learn)

For your AI/ML major project specialization, highlight these key technical points:

• **Model Architecture:** Supervised RandomForestClassifier(n_estimators=40, max_depth=6, class_weight='balanced').
• **Feature Vector:** 5 extracted behavioral features:
1. events_per_sec: Instantaneous I/O velocity.
2. rename_rate: Renaming operations per second.
3. ext_change_ratio: Proportion of file operations targeting suspicious extensions.
4. mod_burst_rate: Rapid write modifications per second.
5. unique_ext_count: Diversity of file extensions impacted.
• **Explainability & Feature Importance:** Feature importance weights are calculated via Gini impurity reduction and rendered in real time on the Settings page.
• **Academic Honesty Note:** Clearly state that the model is trained on 1,500 synthetic behavioral vectors for safe educational demonstration, proving the methodology without requiring live malware detonation.

6. Safety Guarantees (Addressing Faculty Safety Concerns)

Teachers often ask: *'Is this safe to run on university computers? Will it harm my files?'* Answer with these 4 safety guarantees:

• **Strict Sandbox Confinement:** All file operations are programmatically locked to demo_data/sandbox/ via is_safe_sandbox_path(). Path traversal (../) is strictly blocked.

• **Harmless Dummy Files:** Only harmless synthetic text files (e.g. annual_budget.xlsx.txt) are created. No real encryption ciphers or OS-locking code is executed.

• **No Antivirus Interference:** Windows Defender, corporate antivirus, firewall rules, and volume shadow copies (VSS) are never disabled.

• **Zero Private Content Inspection:** Python Watchdog inspects only file event metadata (file name and operation type); file contents are never read, preserving privacy and performance.

7. Top 10 Tough Viva Questions & Perfect Model Answers

Q1: What is the fundamental difference between signature detection and behavioral detection?

Model Answer: Signature detection matches known file hashes (like MD5 or SHA-256) against a static database. It fails against polymorphic malware and zero-day threats where no signature exists. Behavioral detection monitors what the software actually *does* at runtime (rapid renaming, high I/O velocity, extension tampering), allowing us to stop brand-new ransomware variants before they complete encryption.

Q2: How does the sliding time window prevent false positives from everyday user activity?

Model Answer: Everyday user actions (saving a Word document, renaming a spreadsheet) occur at human speeds (1 event every several seconds). Our sliding window evaluates instantaneous velocity over a 5-second window. Unless a process executes multiple renames and modifications per second, the threat score remains strictly in the Low tier (< 30).

Q3: What happens if ransomware encrypts files slowly to evade the rate threshold (Stealth Ransomware)?

Model Answer: This is known as a 'low-and-slow' evasion attack. Our simulator includes a dedicated 'Slow Stealth Simulation' profile. Even if the velocity is low, our multi-factor scoring evaluates extension tampering independently. A single file renamed to a known ransomware extension immediately contributes 28 to 40 points, elevating the incident to Medium or High severity regardless of slow velocity.

Q4: Why did you use Random Forest instead of a Deep Neural Network?

Model Answer: For behavioral telemetry with tabular feature vectors (velocity, rename rate, extension ratios), Random Forest offers superior explainability, requires negligible inference latency (< 5ms), avoids overfitting on small datasets, and provides transparent feature importances via Gini impurity. In cybersecurity EDRs, explainability is essential for SOC analysts to understand why an alert fired.

Q5: What is the difference between Crypto Ransomware and Locker Ransomware?

Model Answer: Crypto ransomware silently searches for valuable personal or business files (documents, images, databases) and encrypts their contents using strong hybrid ciphers (e.g. AES-256 + RSA-4096), demanding payment for the private key. Locker ransomware locks the user interface or boots into a fake lockscreen without necessarily encrypting files on disk.

Q6: How does our automated mitigation mechanism respond to an attack?

Model Answer: When an incident reaches a Critical threat score (≥ 80), our detection coordinator triggers an automated response hook. In this simulation, it automatically suspends the active simulation thread. In an enterprise EDR, this corresponds to terminating the suspicious process tree via API (such as TerminateProcess) and isolating the network interface.

Q7: How does Watchdog monitor file activity without reading file contents?

Model Answer: Python Watchdog hooks into native operating system APIs — specifically the ReadDirectoryChangesW API on Windows and inotify on Linux. The OS kernel notifies Watchdog of file system events (created, modified, deleted, moved) as lightweight metadata notifications. This ensures zero performance degradation and total user privacy.

Q8: What is the 3-2-1 backup strategy in ransomware defense?

Model Answer: 3 copies of critical organizational data, stored across 2 different storage media types (e.g., local NVMe drive and magnetic tape/NAS), with at least 1 copy stored off-site, air-gapped, or in an immutable cloud bucket (WORM - Write Once, Read Many). Even if ransomware encrypts the entire local network, immutable backups guarantee complete recovery without paying a ransom.

Q9: Can an attacker bypass this system by keeping the original file extension?

Model Answer: Some advanced ransomware strains encrypt files in-place without altering the file extension to evade extension-based rules. Our system defends against this through our Burst Modification Rate rule (S_{burst}), which flags rapid overwrites of multiple files within seconds even if extensions remain unchanged.

Q10: What are the main limitations and future scope of this project?

Model Answer: Current limitations: As a user-space Python application, it monitors directory events rather than intercepting kernel I/O requests before they hit the disk. Future scope includes developing a Windows File System Minifilter Driver (kernel-level) to block write operations synchronously, integrating Shannon entropy calculation to detect encrypted blocks before renaming, and mapping detection alerts to the MITRE ATT&CK; framework.